

ULKA JAMBHULKAR

AI Architect

AI Technical Lead with 5+ years of experience delivering AI-driven solutions across agentic AI automation, Generative AI, NLP, Deep Learning, and Machine Learning. Specialized in developing end-to-end AI lifecycles—aligning business strategy with scalable system architectures. Proven ability to accelerate organizational AI adoption through robust architecture, efficient execution, and continuous stakeholder engagement. Keen to stay updated with the latest advancements in the AI & Data Science to drive continuous professional growth.



WORK EXPERIENCE

Chetu, Inc.

OCT 2023 - PRESENT

Multi-Agent Travel Itinerary AI Planner

- Designed and orchestrated a LangGraph-based multi-agent AI automation system for end-to-end trip planning, leveraging OpenAI.
- Automated complex travel planning workflows for city selection, flights, hotels, activities, booking, reducing manual planning effort by ~60%.
- Built independent AI agents with clearly defined responsibilities, seamlessly integrated through controlled execution flow to deliver scalable and reliable automation.
- Implemented agent-to-agent communication and shared state management using LangGraph to ensure coherent, context-aware trip generation.
- Integrated REST APIs to enrich agent outputs with their preferences.
- Applied prompt engineering, structured outputs, and validation logic to ensure accuracy and consistency across automated workflows.

GenAI Automation for Visual Content and Architecture Diagrams

- Built an AI-driven automation platform that generates PowerPoint presentations, short videos, flyers, and visual assets directly from prompts using OpenAI GPT for content generation and Gemini API for image creation via diffusion models.
- Automated architecture diagram generation from natural language prompts using PlantUML, covering Use Case, Class, ER, Activity, and Architecture diagrams within a single unified workflow.
- Applied advanced prompt engineering techniques to extract structured components from user inputs, ensuring consistent diagram generation.
- Enabled editable, API-integrated UI workflows, allowing internal teams to modify generated diagrams and content, reducing manual design and documentation effort by ~40% through end-to-end GenAI automation.

AI Automated Hiring & Promotion Decision Platform

- Built an end-to-end AI automation platform for candidate communication assessment using OpenAI GPT models, reducing manual evaluation effort by ~50% for the hiring team.
- Automated listening comprehension, grammar, fluency, semantic analysis, and response evaluation, with speech-to-text processing using OpenAI Whisper for accurate transcripts.
- Developed a Promotion Module with 4 AI-driven assessment areas: Comprehension Retention, Situational Conversation, Emotion Detection, and Email Writing, enabling the system to select the best candidates for promotion and bring versatility to promotion decision-making.
- Mapped candidate performance to CEFR proficiency levels and integrated promotion metrics, enabling standardized and AI-driven HR decisions, reducing manual effort by ~40% for internal HR teams.

CONTACT

Phone

8510802840

Email

ulkajambhulkar280@gmail.com

Address

Amrapali Dream Valley, Gautam Buddha Nagar, Techzone -4, Greater Noida

PROFESSIONAL SKILLS

- Project Management
- Public Relations & Executive Stakeholder Engagement
- Leadership, Team Development & Interview Panelist
- Strategic Planning & Decision-Making
- Time Management & Prioritization
- Cross-functional Collaboration
- Communication & Presentation Skills

TECHNICAL SKILLS

Agentic AI: Multi-Agent AI Systems, AI Automation, LangGraph, Phidata, CrewAI, AutoGen, Agent Orchestration

Generative AI: Large Language Models ((OpenAI, LLaMA, Mistral), Prompt Engineering, Retrieval-Augmented Generation (RAG), Fine-tuning (SFT, Instruction Tuning, LoRA, QLoRA), Hugging Face, Transformers, LangChain, LlamaIndex, Vector Databases, Diffusion Models

PAINT BY NUMBER – CONTOUR ANNOTATION

- Developed an AI-driven Paint by Number system to automatically convert images into numbered color palettes using Python, OpenCV, scikit-image, PIL, and KMeans clustering.
- Implemented color clustering, HEX/RGB conversions, palette matching, and Canny edge detection algorithm to identify dominant colors and map them to a predefined palette with accurate color IDs.
- Generated faceted masks for each color, enabling precise segmentation and clean color assignments.
- Automated isolated pixel correction, narrow strip cleanup, and small contour merging, ensuring clean, visually coherent Paint by Number outputs.
- Generated contour-based numbered annotations and exported final images in both PNG and SVG formats for versatile usage and printing.

RAG-based RFP Generation & Automation

- Developed an RAG system for automated Request for proposal creation, integrating organization documents into a cohesive knowledge base.
- Utilized Qdrant vector database for storing embeddings of document sections and implemented semantic search for efficient retrieval of relevant content.
- Processed text data with tokenization and embedding techniques, enabling accurate similarity matching and content extraction across diverse document types.
- Applied OpenAI LLM for summarization and synthesis of retrieved content, generating a structured analysis summary from multiple RFP sections.
- Built a FastAPI to expose the RFP generation workflow, allowing frontend teams to fetch preprocessed content and assemble complete RFPs in doc format.
- Ensured end-to-end automation from document ingestion, embedding, retrieval, summarization, to RFP generation, improving accuracy by finetuning, speed, and consistency in organizational proposals.

Foster Technologies India

Dec 2020 - Oct 2023

ML-Based Fuel Delivery Automation & Optimization

- Integrated Python and .NET APIs to fetch and preprocess real-time tank, vendor, vehicle, and driver data using Pandas, handling parameters like tank burn rate, trailer loading time, and refinery distance.
- Implemented Priority-based refuelling for tanks (<30% capacity) with real-time data, Haversine distance calculations, and structured Pandas DataFrames for decision-making.
- Optimized delivery logistics using K-Means clustering and GraphHopper API for route optimization to group tanks by refinery and assign nearest vehicles with sufficient fuel volume.
- Automated vehicle-driver assignment considering driver licenses, certifications, trailer type, capacity, and proximity, ensuring efficient and compliant fuel distribution.
- Calculated trip schedules, delivery time, and fuel requirements using GraphHopper and other route optimization APIs, enabling end-to-end route planning.
- Developed scalable pipelines for fleet management, vendor mapping, and real-time monitoring using Pandas, improving operational efficiency and delivery accuracy.

CERTIFICATIONS

- IBM Data Science Professional
- Certificate of “Basics of Remote Sensing, Digital Image Processing, GIS” by National Remote Sensing Centre, ISRO - Nagpur

Natural Language Processing (NLP):

Transformers, BERT, LLaMA, GPT-based models, Sentiment Analysis, Text Classification, Summarization

Deep Learning & Computer Vision:

CNNs, OpenCV, Image Classification, Object Detection, Image Segmentation, Transfer Learning, Vision Transformers

Machine Learning: Supervised &

Unsupervised Learning algorithms, Feature Engineering, Model Training & Optimization

Data Engineering & Pipelines:

Pandas, NumPy, SQL, GenAI data ingestion & ETL, RAG pipeline design, vector indexing & semantic retrieval using FAISS, Pinecone, and ChromaDB.

Programming & Frameworks:

Python, PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face (Transformers, Tokenizers), OpenCV, spaCy, NLTK, NumPy, Pandas

Model Deployment:

FastAPI, Flask, CI/CD pipelines, AWS, Azure ML, On-premise AI orchestration, Monitoring & Scaling

Databases:

PostgreSQL, MySQL, Qdrant, FAISS, Chroma

AI-assisted Development Tools :

ChatGPT, GitHub Copilot, Google Colab, VS Code

LANGUAGES

- English
- Hindi
- Marathi

HOBBIES

- Drawing Mandala & Zentangle Arts
- Cooking
- Gardening